

FORECAST BUST SENTINEL

AI-Based Forecast Reliability Intelligence for Medium-Range Forecasts

A reliability layer over an existing NWP ensemble: it estimates where and when (Day 1–10) the forecast is likely to incur a large error — it does not replace the forecast.

Problem Statement ID – SIH26079

Problem Statement Title – AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts

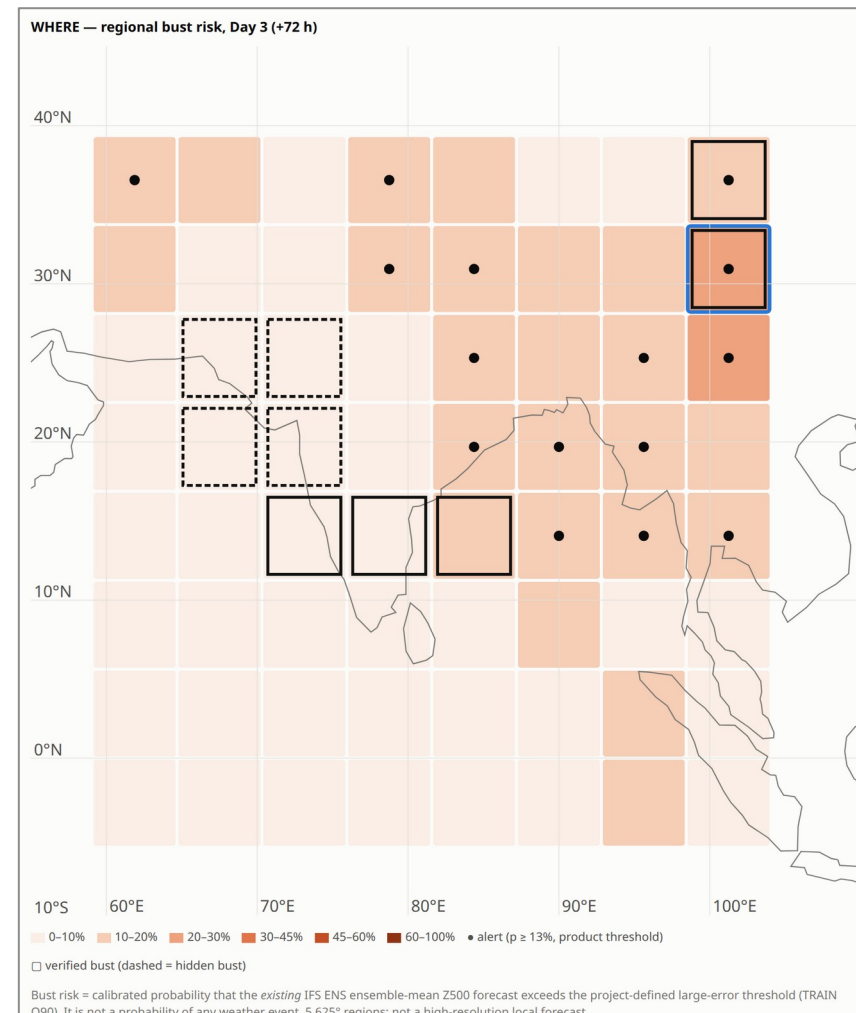
Organization – Ministry of Earth Sciences (MoES) — National Centre for Medium Range Weather Forecasting (NCMRWF)

Theme – Smart Automation

PS Category – Software

Team ID – 174190

Team Name (Registered on portal) – OVER_CLOCKED



Real prototype output (historical replay, 2022-02-02 12 UTC): calibrated Day 3 bust risk over 64 native 5.625° regions (ECMWF IFS ENS via WeatherBench 2); outlined boxes = ERA5-verified busts after reveal.

We do not replace the NWP forecast — we estimate when it is likely to fail, show why, and learn from verified failures

The Forecast Reliability Challenge

Busts in rapidly evolving systems

Monsoon depressions, WDs, cyclones, heat waves, active/break phases → large medium-range errors that affect operational decisions.

Ensemble spread helps — but is not enough

IFS ENS spread–error correlation only 0.21 (Day 1) to 0.56 (Day 10) over our target regions (2022).

Region × lead-time reliability gap

Forecasters need WHERE and WHEN to be cautious, per region, Day 1–10 — not one global skill score.

History as evidence

Verified errors of similar past forecast states are rarely available at forecast time.

How Forecast Bust Sentinel Works

01

CHARACTERIZE

Current NWP forecast + ensemble behaviour + atmospheric state

02

PREDICT

Calibrated bust probability per region, Day 1–10

03

CONTEXTUALIZE

Compare with strong spread-only baseline (B2) + retrieve similar historical forecast states

04

QUALIFY

Support/OOD + evidence strength + attribution + forecaster review priority

05

VERIFY & REMEMBER

ERA5 verification + failure fingerprint + expanding historical memory

Project-Level Differentiation

Conditional Reliability Layer

“What does the current forecast state imply about its reliability?”

Historical Forecast-State Memory

“What happened when previous forecasts looked like this?”

Spread-vs-Sentinel Disagreement

“What does the richer model see that spread alone does not — and vice versa?”

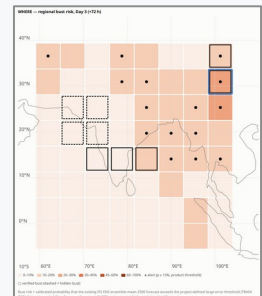
Evidence-Aware Reliability

“Is the risk strongly supported or based on weak precedent?”

Closed Verification Loop

Forecast → verification → failure fingerprint → reusable memory

Live prototype evidence (historical replay, real 2022 IFS ENS case)



Regional Day 3 risk map

Forecaster review priority						
#	Region	Day	Bust risk	B2	Δ pp	Evidence
1	R0607 Himalaya / Tibetan Plateau	D3	25%	25%	0	(WEAK) ✓ bust
2	R0507 Myanmar / Indochina	D3	24%	24%	0	(WEAK) ✗ no bust
3	R0305 Bay of Bengal	D3	20%	20%	0	(MODERATE) ✗ no bust
4	R0307 Myanmar / Indochina	D3	16%	16%	0	(STRONG) ✗ no bust
5	R0303 South Indian Peninsula	D2	16%	16%	0	(MODERATE) ✗ no bust
6	R0304 Bay of Bengal	D2	16%	16%	0	(MODERATE) ✗ no bust
7	R0203 South Indian Peninsula	D5	20%	20%	0	(STRONG) ✗ no bust
8	R0305 Bay of Bengal	D2	15%	15%	0	(MODERATE) ✗ no bust
9	R0503 Central India	D2	25%	25%	0	(INSUFFICIENT HISTORICAL SUPPORT) ✓ bust
10	R0004 Equatorial Indian Ocean	D2	12%	12%	0	(STRONG) ✗ no bust
11	R0602 Northwest India / Pakistan	D2	24%	24%	0	(INSUFFICIENT HISTORICAL SUPPORT) ✓ bust
12	R0203 South Indian Peninsula	D6	25%	25%	0	(MODERATE) ✗ no bust

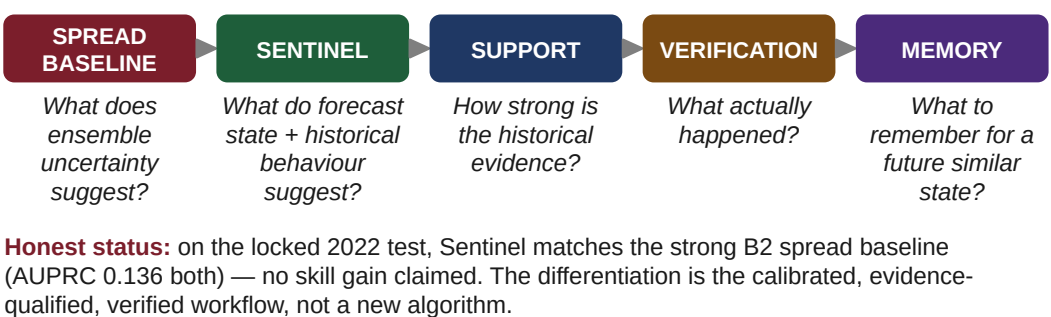
Review priority + B2 + support, after reveal

WHY — evidence for R0607, Day 3

- A. Ensemble evidence.** Z500 ensemble spread 11.8 m = 100th percentile of training spread for this region, lead day and season; member anomaly-sign agreement 100%.
- B. Atmospheric-state evidence.** Forecast Z500 anomaly -46 m ([anomaly] at the 94th training percentile); pattern distance from the training mean 3.94 (PC units).
- C. Historical evidence.** Of the 30 most similar verified historical forecast states for this region and lead day (1492 verified cases available; 7 within the similarity radius), 3% exceeded the bust threshold (climatological rate 10%).
- D. Forecast-evolution evidence.** Versus the cycle 24 h earlier (same valid time) the regional Z500 forecast moved 2.0 m (neighbourhood RMS revision 2.6 m, 45th training percentile).
- C2. Recent verified error behaviour.** Over the 5 days before initialisation, verified Day 1-3 forecasts for this region had mean normalized error 0.15 (mean error +4.6 m; 0% busts, n = 14); neighbouring regions 0.15, whole domain 0.12.
- E. Baseline disagreement.** Sentinel 25% vs calibrated spread-only baseline 25% (+0 percentage points).

WHY panel: number-backed evidence A–E

Where the differentiation lies — the integrated workflow



From a real 50-member ensemble to a calibrated, verified, explainable Day 1–10 reliability output

1 DATA & VERIFICATION

ECMWF IFS ENS
50 members, 2018–22
(WeatherBench 2)

ERA5 reference
at valid time
(init + lead)

Area-weighted
regional Z500
RMSE, Day 1–10

Historical
error archive
(verified cases)

- Grid: native 64×32 (5.625°) product — 64 target regions over 5°S–40°N, 60–105°E; wider context domain for features. Not 5°×5° (bandwidth-limited download).
- ERA5 is a reanalysis reference, not perfect truth. Anomaly climatology from ERA5 1990–2017 (pre-dates all experiment years).

2 TARGET & MODELLING

Region ×
Day 1–10

Regional
Z500 RMSE

TRAIN-derived
normalisation

TRAIN-only Q90
(region·lead·season)

Bust
label

B0
Climatology
TRAIN bust frequency

B1
Spread
spread-percentile
score

B2
Calibrated spread-only
GBT + isotonic (strong)

Sentinel
Forecast-state model
one shared XGBoost (CPU hist): region,
lead, season + state features

3 FEATURE INTELLIGENCE (forecast-initialisation info only)

ENSEMBLE BEHAVIOUR
spread, percentile, IQR, skew, member agreement

LARGE-SCALE PATTERN
TRAIN-only PCA, 8 frozen components

HISTORICAL MEMORY
analogues (strictly earlier inits) + recent verified error

ATMOSPHERIC STATE
Z500/700/850 anomaly & gradients (wind/MSLP: dev experiment, pending)

FORECAST EVOLUTION
revision vs earlier cycle, same valid time

Rule fixed in advance: a group enters Sentinel only if it beats B2 on 2021 validation. None did → FULL uses B2 inputs.

4 VALIDATION & TRUST — locked 2022 test, scored once

Train 2018–20 (1,096 inits)

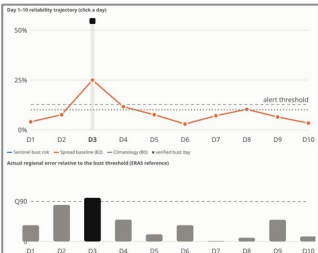
Validation 2021 (364)

Locked test 2022 (366)

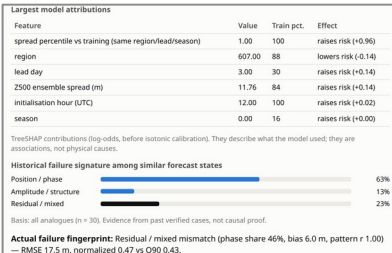
	2022 test	AUPRC	ROC AUC	Brier	ECE	Rec@10%FAR	Hidden-bust rec.
B0 climatology		0.091	0.517	0.0804	0.0134	0.075	0.073
B1 spread score		0.131	0.609	—	—	0.195	0.000
B2 calibrated spread		0.136	0.609	0.0791	0.0072	0.198	0.000
Sentinel (FULL)		0.136	0.609	0.0791	0.0072	0.198	0.000

Test base rate 0.088 · 234,240 region×day cases · warning lead of detected busts 6.3 d · spatial Jaccard 0.082 · spread-skill ratio 0.79 → ~1.0

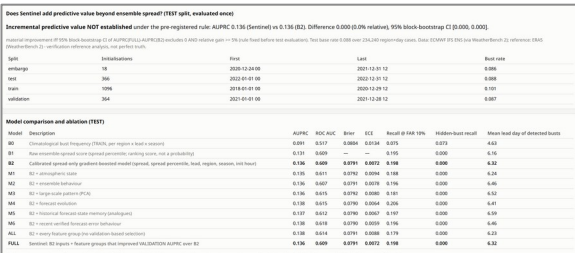
5 PROTOTYPE: FastAPI + React dashboard on precomputed real 2022 replay cases



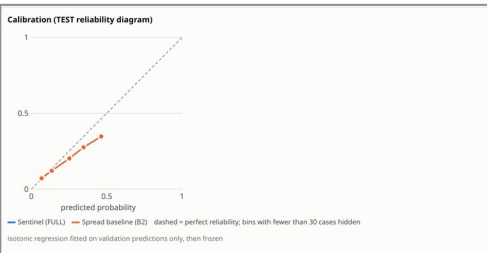
Day 1–10 reliability trajectory vs B2 + verified error



Largest model attributions



Model comparison and ablation (TEST)



Calibration (TEST reliability diagram)

Stack

- Python · xarray/Zarr
- XGBoost (CPU) · isotonic
- TreeSHAP attribution
- FastAPI · React/Vite
- 46 pytest + 3 vitest pass

1 FEASIBILITY — demonstrated

- Real data: 5 years of 50-member ECMWF IFS ENS + ERA5 (WeatherBench 2), 457/457 chunks downloaded (~4.5 GB).
- CPU-friendly: one shared gradient-boosted model, no GPU; full train → evaluate → replay → UI via scripts/run_all.sh.
- Modular ingestion: ForecastSourceAdapter interface (IFS ENS adapter live; NCMRWF boundary defined).
- Reproducible: manifests, frozen thresholds/PCA/calibrators, seeds, software versions persisted.
- Precomputed replay: 12 real 2022 cases, blind forecast file + separately served verification.

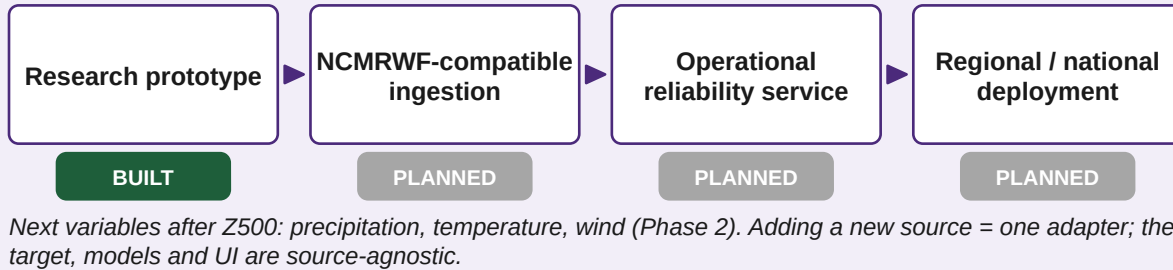
3 MITIGATION & TRUST

- Chronological split + purged embargo; train-only normalisation, thresholds and PCA; validation-only calibration.
- Analogue memory uses strictly earlier initialisations; pre-registered improvement rule; 2022 scored once.
- Support/OOD category + evidence strength shown next to every probability; weak support is visibly flagged.
- Full provenance in the UI; forecaster makes the call — Sentinel only prioritises review.

2 KEY CHALLENGES — stated honestly

- No NCMRWF NEPS data accessible yet → not integrated or evaluated on NCMRWF output.
- Variables: geopotential only so far; wind/MSLP/vorticity features in a dev-split experiment (pending).
- Predictability ceiling: on 2022, Sentinel did not beat spread-only B2; hidden busts (low spread) not detected.
- Weak historical support for novel states; one test year; research archive ≠ operational performance.
- Resolution: native 5.625° boxes (~1 MB/s link), not 5°×5° or high-resolution local output.

4 VIABILITY & SCALE — path



IMPLEMENTATION STATUS — current vs planned

Implemented: data pipeline · labels · B0/B1/B2 · Sentinel + ablations · calibration · analogues · support/OOD · attribution · replay + verification · fingerprint · API · dashboard. **In progress:** wind/MSLP dynamics group (dev split only). **Planned:** NCMRWF ingestion, live operation.

Helping forecasters answer: where to look, when to be cautious, why, how strong the evidence is — and what happened

1 KEY USERS

- NCMRWF medium-range forecasters (primary)
- Weather-sensitive operational decision makers
- Disaster-management stakeholders — via forecaster-reviewed guidance, not public warnings

2 OPERATIONAL BENEFITS

- Earlier identification of regions where the forecast may be unreliable
- Lead-time-aware: Day 1–10 trajectory + peak-risk day
- Reduced manual screening: transparent review-priority queue
- Transparent, number-backed reasons

3 FORECAST KNOWLEDGE

- Historical forecast-state memory grows with every verified forecast
- Failure fingerprints (position/phase, amplitude/structure, residual)
- Verification feedback per region and lead day
- Reusable, auditable evidence

4 STRATEGIC VALUE

- Better interpretation of medium-range forecast reliability
- Supports weather-sensitive planning with qualified confidence
- Source-agnostic architecture that can sit on NCMRWF ensembles (planned)
- No lives-saved or savings figures claimed — not yet measured

REAL-WORLD IMPACT SCENARIO — replayed on a real 2022 case (2022-02-02 12 UTC, hidden-bust stress case)

1

Forecast issued
IFS ENS 50-member run; Sentinel sees only init-time information (blind mode)

2

Reliability screened
64 regions × Day 1–10 scored; calibrated risk vs B2 vs climatology

3

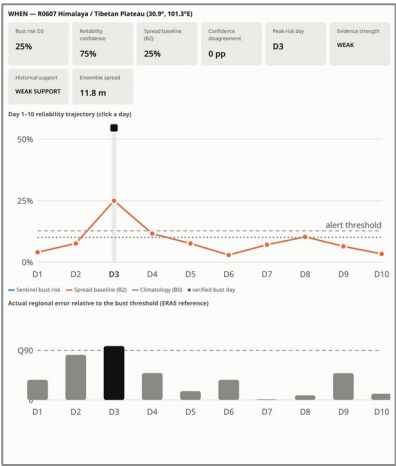
Risky region identified
R0607 Himalaya/Tibetan Plateau, Day 3: 25% risk (climatology 10%), evidence WEAK

4

Forecaster investigates
Evidence A–E: spread at 100th pct, anomaly, 30 analogues, revision, recent errors

5

Verified + memory updated
ERA5 reveal: bust at Day 3; fingerprint residual/mixed stored for future analogues



Same case in the prototype after “Reveal verification” (real replay output)

Research Sources

SIH26079 problem statement — MoES / NCMRWF, Smart India Hackathon 2026.

Rasp, S. et al. (2024) WeatherBench 2: A benchmark for the next generation of data-driven global weather models. JAMES 16, e2023MS004019. (IFS ENS + ERA5 archive used)

Hersbach, H. et al. (2020) The ERA5 global reanalysis. Q. J. R. Meteorol. Soc. 146, 1999–2049.

Bougeault, P. et al. (2010) The THORPEX Interactive Grand Global Ensemble (TIGGE). Bull. Amer. Meteor. Soc. 91, 1059–1072.

Leutbecher, M. & Palmer, T. N. (2008) Ensemble forecasting. J. Comput. Phys. 227, 3515–3539. (spread–skill)

Rodwell, M. J. et al. (2013) Characteristics of occasional poor medium-range weather forecasts for Europe. Bull. Amer. Meteor. Soc. 94, 1393–1405. (forecast busts)

Chen, T. & Guestrin, C. (2016) XGBoost: A scalable tree boosting system. Proc. KDD 2016, 785–794.

Lundberg, S. M. et al. (2020) From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence 2, 56–67. (TreeSHAP)

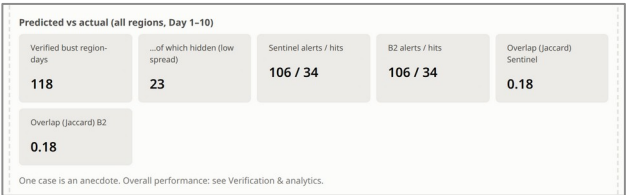
Links & Resources

GitHub repository: github.com/Vishant-1001/hmmm

Prototype: runs locally — scripts/serve.sh (FastAPI + built UI); no public deployment yet

Docs in repo: methodology · leakage controls · evaluation · model card · limitations

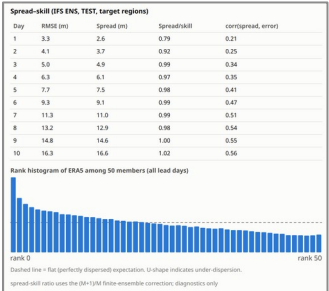
Prototype Evidence (real 2022 replay / test artifacts)



Verification reveal: predicted vs actual busts, all regions, Day 1–10 (one case)

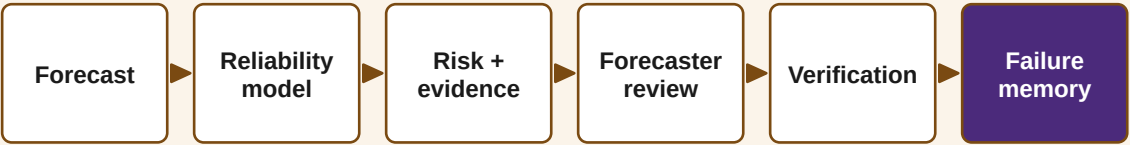
Diagnostics			
Peak-risk day error (Sentinel, mean Δday)	3.07 (within ±1 day: 0.35)		
Peak-risk day error (B2)	3.07		
Spatial overlap, mean Jaccard (Sentinel / B2)	0.082 / 0.082		
Low-spread AUPRC (Sentinel / B2)	0.060 / 0.060		
Q95 sensitivity AUPRC (FULL / B2 / B0)	0.084 / 0.084 / 0.042		
Frozen-memory FULL AUPRC	0.136		
Failure-signature top-1 agreement (analogues / climatological reference)	0.409 / 0.332		
Mean probability assigned to actual signature (analogues / reference)	0.384 / 0.337		
Evidence strength vs outcome (TEST)			
Evidence	n	Mean Sentinel p	Observed bust rate
STRONG EVIDENCE	178485	0.090	0.081
MODERATE EVIDENCE	16231	0.100	0.113
WEAK EVIDENCE	11380	0.092	0.108
INSUFFICIENT HISTORICAL SUPPORT	28144	0.095	0.110

Diagnostics incl. failure-signature agreement 0.409 vs 0.332



Spread-skill + ERA5 rank histogram (2022)

Decision flow



Failure memory feeds the analogue search for the next forecast (loop closed in replay; live operation planned).